

SCI-MAP

Ordinary Mapmaking and Observation Coaddition

Engineering Conformance Specification

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1 Purpose and conformance target

This specification tells a reviewer how to decide whether one exact candidate conforms to SCI-MAP v0.2. It does not select a candidate, inspect an implementation, execute validation, or assert a result. Before work begins, the reviewer freezes:

- repository and immutable candidate identity;
- build profile, compiler/runtime identities, configuration, and enabled parallel modes;
- exact input fixtures and independent expected-value generators;
- numerical comparison policy and simultaneous statistical error policy;
- required product list, representation paths, log policy, and completion criteria; and
- the shared-authority digest, contract version, this document revision, owner-decision ledger revision, exact byte-identical PTC-to-MAP boundary digest, exact original-footprint-coordinate boundary, source-bound MAP profile versions, six product-role identities, coadd Registry record, owner r0.2 directive identity/digest, and the exact candidate paired `SCI-VAL_PROFILE_REGISTRY v0.1/r0.3-map-v0.2-r0.2-candidate-2026-09-07` and `SCI-VAL_SOURCE_BINDING_REGISTER v0.1/r0.3-map-v0.2-r0.2-candidate-2026-09-07` records. Candidate records are not treated as owner-accepted, canonically activated, or frozen merely because their bytes and hashes are complete.

Evidence from a different candidate, configuration, or science-authority digest is not silently pooled. Each rerun keeps its own identity and reason. Historical operational acceptance does not substitute for conformance evidence.

The exact MAP lifecycle is **requested**, **effective**, **observation_resolved**, **applied**, **realized**, in that order. **requested** is immutable accepted intent. **effective** is its supported executable interpretation. **observation_resolved** means every observation-specific parent, coefficient, profile, geometry, mapping, parameter, support rule, output role, and persistence decision is concrete or typed unavailable. **applied** binds the exact resolved operation and parameter state to the invocation or commitment of one execution attempt. **realized** binds the immutable resulting accumulators, products, statuses, failures, and provenance that were actually produced. A stop before application has neither an applied nor a realized record. An applied attempt that terminates before output realization retains its applied-attempt identity and exact terminal failure but creates no realized product.

MAP stage	Exact meaning	Record rule
requested	Immutable accepted intent.	Exists only for an accepted request; it is never reconstructed from a later stage.
effective	Supported executable interpretation of that request.	Binds supported/disabled/unavailable interpretation without mutating intent.
observation_resolved	Every observation-specific parent, coefficient, profile, geometry, mapping, parameter, support rule, output role, and persistence decision is concrete or typed unavailable.	Supplies the exact value/state for pre-application validation.
applied	Exact resolved operation and parameter state was invoked or committed for one execution attempt.	Created only at the application event; retained with exact terminal failure when output realization fails.
realized	Immutable resulting accumulators, products, statuses, failures, and provenance were actually produced.	Created only for actual realization; it never authorizes its own earlier application.
Termination rule	Stop before application, or stop after application but before output realization.	The former has neither applied nor realized record; the latter has an applied-attempt record and terminal failure but no fabricated realized product.

The MAP plan, coefficient selection/application, projection, support policy, output/coadd plan, response/covariance request, product role, and **coverage_cut** use this same ordering. Each parameter stage binds its exact value and provenance; a completed realization contains all five stages, while an earlier stop contains only stages actually reached.

The MAP lifecycle does not replace the independently named PTC/VAL handoff axes **requested/not_requested**, **applicable/inapplicable/applicability_unknown**, **eligible/ineligible/decision_unavailable**, and **realized/incomplete/failed/not_produced**. It also does not replace the ECS requirement-result vocabulary or evidence- artifact realization state. In particular, a MAP parameter's resolved or applied value may be checked

before application or bound after application; its **realized** value is never a prerequisite for a pre-application check.

2 Evidence record

Each requirement record contains at least:

Field	Required content
Requirement	Stable SCI-MAP-REQ-NNN and shared-authority digest.
Candidate	Exact source/build identity and environment profile.
Observable	Public product, typed state, API boundary, metadata, diagnostic, or failure transition under test.
Fixture	Immutable input identity and independent expected-value calculation.
Comparison	Exact predicate or preregistered numerical/statistical bound.
Result	Exactly one of pass , fail , blocked , not_applicable , or not_assessed . Skipped required evidence is not_assessed , never a pass.
Artifact realization	Exactly one of complete , incomplete , failed , or not_produced ; this records the evidence artifact and never substitutes for the requirement result.
Artifacts	Machine-readable outputs, logs, traces, plots/tables where needed, and digest-addressed provenance.
Reviewer	Independent reviewer identity, date, and deviation disposition.

Every ECS requirement result is exactly one of **pass**, **fail**, **blocked**, **not_applicable**, or **not_assessed**. **pass** means complete positive evidence establishes the requirement for the exact candidate. **fail** means evaluated evidence contradicts it. **blocked** means evaluation cannot proceed because one exact named authority, dependency, profile, source binding, or prerequisite is unavailable. **not_applicable** means the requirement is outside the exact candidate's declared role under an owner-approved rationale. **not_assessed** means applicable evidence has not yet been executed or reviewed. Evidence-artifact realization is a different axis with values **complete**, **incomplete**, **failed**, and **not_produced**.

Failures are retained as evidence. A tolerance, fixture, expected result, or applicability decision is not changed after viewing a failure without opening a new preregistered record.

3 Claim-layer gates

Passing one layer grants no stronger layer automatically.

Layer	Minimum engineering record	Prohibited inference
Algebraic contract	Exact owner-dispositioned shared authority and internally consistent independent expected values.	No claim that a candidate implements it.
Engineering conformance	Every applicable requirement passes on one exact candidate; no required skip, unresolved deviation, or false completion.	No claim that persisted WCS/response is scientifically faithful in real data.
Representation/response fidelity	Round-trip and injection evidence covers WCS, units, response classes, boundaries, and covariance representations with preregistered bounds.	No observational photometric, astrometric, or noise performance claim.
Observational performance	Independent references and repeatability establish the named amplitude, shape, position, covariance, or false-rate metric.	No production-readiness claim.
Production readiness	Operational profile, complete products, failure/log audit, dependency closure, release controls, and owner approval all pass.	No authority beyond the exact released profile and contract version.

All nine stable MAP scope decisions are decided for this candidate. A claim may still be blocked by an unapproved source binding, an unavailable producer, or missing conformance evidence; record that state without weakening the requirement or substituting a fallback.

4 Execution order and atomicity protocol

Run evidence in the following order so later evidence cannot mask an earlier contract failure:

- 1. Stage A—structural.** Parse and freeze exact identity and parent, generation, source/profile binding, producer availability, and required decision realization for frozen PTC r0.5, the exact PTC-to-MAP boundary, and the AST same-*n* signal coordinate, all boundaries/profiles, units, lifecycle, WCS, MAP policy, and required-product plan. When the uniform route is selected, require the exact uniform Registry, family, and handoff identities declared in the shared authority; selection alone supplies no coefficient and creates no default. Reject direct CAL, PTC-disabled, missing-product, inferred no-op, and cross-generation routes before payload retrieval.
- 2. Stage B—safe numerical classification.** Retrieve only structurally admitted transformed-signal and coefficient payloads. Classify z_i as finite, non-finite, or unrepresentable and γ_i as finite strictly positive, exact zero, negative finite, non-finite, or unrepresentable. Verify coefficient availability itself imposed no finiteness requirement. Separately validate the exact observation-resolved coverage cut before application: zero, ordinary (0,1], above one admitted only by an independently authorized same-value/scope `coverage_cut_above_one_override_requested` fact, or invalid when missing, non-finite, negative, or unrepresentable. The number above one never authorizes itself. Invalid input fails before support construction, application, or mutation; no realized cut value is a pre-application prerequisite.
- 3. Stage C—coordinate and product.** Evaluate exact AST coordinate validity, one-hot placement/boundary, and only the companions required by the exact selected observation/coadd product role. For every VAL-evaluated gate retain request, applicability, eligibility, and realization axes, source binding, immutable reason, cause, and failure scope. Only requested/applicable/eligible/realized passes. Verify separately that the PTC coefficient/QC pass uses its exact PTC-owned profile and neither replaces nor rescues `SCI-MAP:map_upstream_admission@2`.

4. **Stages D and E—membership then accumulation.** Construct membership only after Stages A–C and perform no payload arithmetic until all required gates pass. Verify no numeric-zero mask, late masking, or aggregate sanitization substitutes for admission. Then compare exact scientific output-row identity and cardinality, plus one-pixel, one-hot half-open-boundary, prohibited-fractional, covariance, fixed-state response, full-procedure comparison-space typing, and distinct-fact observation outputs, to independent expected values; retained storage rows do not count as scientific output rows.
5. Exercise original-footprint exposure fixtures: one original influencing several descendants at different pixels; original and descendants in different pixels; missing original coordinate; boundary loss; multiple maps sharing one original; distinct observations with numerically identical original coordinates; and donor/synthesized descendants. Verify exact observation/UID/native/stable-ALIGN-slot/generation keys, layered AST ALIGN-grid parent in the target WCS, one own-coordinate placement, separate influence edges, and no precision or temporal-support promotion.
6. Verify complete immutable observation-bundle construction as a logical artifact. Independently request all three observation roles and record each reached requested, effective, observation-resolved, applied, and realized plan stage; a completed realization has all five, a pre-application stop has no applied/realized records, and an applied terminal failure has no realized product. Exercise required publication for persisted routes plus explicitly selected in-memory and coadd-only routes. In-memory retention is not a waived product.
7. Snapshot coadd-owned state; test every compatibility failure under each selected role; require atomic nonmutation. Confirm that base-permitted unavailable or basis-incompatible response and incomplete covariance preserve the base signal membership and publish exact status/cause.
8. Evaluate all four axes of the registered observation-coadd admission aggregate profile separately from the MAP lifecycle; then independently request all three coadd roles and exercise admitted centered-integer coaddition, exact coadd output-row identity, and companion placement. If a canonical target grid was requested before observation-map construction, verify that every constructed map honors it. Otherwise verify that already existing compatible bundles, including persisted coadd-only inputs, may select a common coadd target and use exact centered-integer embedding without modifying or relabeling their scientific samples. An explicit crop creates a new row-selected product with its own identity and provenance.
9. Round-trip all required representations and reconstruct each row-selected operator from provenance. Evaluate the physical FITS and typed WCS on every preregistered scientifically addressable pixel center and every outer-footprint cell vertex, including all corners, with the exact index mapping, common WCS generation/frame, binary64 great-circle metric, and fail-on-invalid rule. Infer no continuum maximum from that finite population.
10. Exercise fixed-state, PTC full-procedure, and PTC+MAP re-resolved response families. For every numerical Δz or Δm , require the complete baseline/perturbed identities, quantity/unit, rows, frames/WCS, reference/gauge/null state, supports, coefficient/response bases, exact common domain and both maps, unmatched-row treatment, perturbation parameters, finite-difference convention, state transition, availability, and cause. Reject implicit fill, intersection, coordinate/shape/position association, interpolation, reprojection, or gauge alignment; a state delta alone never authorizes subtraction. Reconstruct and disclose the complete MAP-local operator $A_{\text{MAP},\Pi}$, the immutable realized plan Π , and row selector J_{out} with its exact output-row domain for every valid product, while marking the whole-chain RTC-to-CAL-to-PTC-to-MAP role unavailable unless a separate approved authority establishes it; exercise an already realized PTC-grid companion applied exactly once; and honest response/covariance disclosure. For a response-bearing role, require only every exact response object named by that role; absence of a different unselected response family does not block it. For coadd covariance, require every block named by the selected covariance-qualified role before accepting the complete numerical propagation and otherwise require partial, symbolic, summarized, lineage-resolvable, or unavailable state on the base role. Verify no stronger role subsets an existing base parent and verify versioned binding for later derivative products.
11. Exercise consumer claim boundaries and immutable base/unfiltered MAP parentage without assuming a closed-world registry of all future scientific uses. Verify that Pointing and OOF receive only permission to reuse the exact ordinary MAP gridding operator when their inputs conform. Their WCS/frame, model/fitting, interpretation, outputs, defaults, profiles, and qualification remain mode-owned, and no unsupplied mode profile is activated.
12. Repeat supported execution modes under the preregistered floating-point policy, while requiring exact discrete facts.
13. Only after deterministic gates pass, execute representation, external, observational, and operational evidence appropriate to the requested claim.

For every failure-injection test, capture pre-state, attempted input, emitted diagnostic, post-state, publication/cardinality state, and completion state. No partial-state comparison may omit a coadd companion merely because its payload was not the injected failure target.

4.1 Scope-discrimination cases

The following cases make the targeted r0.2 amendments discriminating. Each case records every reached requested, effective, observation-resolved, applied, and realized state and asserts role-qualified atomic nonmutation on rejection.

The six independently requestable product roles are `SCI-MAP:base_observation_map@1`, `SCI-MAP:response_bearing_observation_map@1`, `SCI-MAP:covariance_qualified_observation_map@1`, `SCI-MAP:base_coadd@1`, `SCI-MAP:response_bearing_coadd@1`, and `SCI-MAP:covariance_qualified_coadd@1`. Every request has its own immutable identity, parent set, ordered MAP lifecycle, result, and failure scope. A stronger role is a new product over the exact base parent; it does not rewrite that parent's signal membership, response/covariance status, validity, or provenance.

Role	Exact parent	Required companions	Missing/incompatible companion
Base observation	Exact realized PTC/AST inputs and complete MAP-local observation plan	Complete MAP-local realized operator identity; honest upstream-response and covariance states	Base signal may remain valid; state and cause are retained.
Response-bearing observation	Exact base-observation product named above	Every named numerical response, with exact source domain, basis, class, unit, normalization, parent, WCS/grid, and row identity	This role fails or is unavailable before its required-product mutation; its base parent is unchanged.
Covariance-qualified observation	Exact base-observation product named above	Every within-observation covariance block named by the role on exact rows	Missing blocks are unknown and block this role only; its base parent is unchanged.
Base coadd	Ordered set of exact complete base-observation bundles and one resolved coadd plan	Every base signal companion and honest response and covariance identities/statuses	Unavailable or basis-incompatible response and incomplete covariance are permitted with unchanged signal membership; coadd response becomes unavailable with exact cause.
Response-bearing coadd	Exact base-coadd parent named above and the identical ordered signal-member set	Every member response and their exact compatible source domain, basis, class, unit, normalization, parent, WCS/grid, and row identity	If the base parent already exists, this stronger role is unavailable for that parent; it never selects a subset. A separately requested admission may reject an observation before mutation only while constructing its own separately identified base/signal parent. No hidden subset or zero fill.
Covariance-qualified coadd	Exact base-coadd parent named above and the identical ordered signal-member set	Every named within- and cross-observation block on exact stacked rows	One missing required block blocks this role for that parent; no member subset is formed. A separately requested admission may reject before constructing its own identified base/signal parent. Unknown is never zero or independence.

Decision surface	Admitted case	Required discriminator
Uniform coefficient authority	Exact Registry/family/handoff selection and a PTC-complete same-generation payload.	Admit only through the frozen PTC-to-MAP boundary; no default, fallback, or coefficient synthesis.

Coverage cut	Finite requests at 0, within (0, 1], and above 1.	Classify zero and ordinary values separately from the independently authorized typed above-one override; reject numerical self-authorization and missing, negative, non-finite, or unrepresentable values before support/application/mutation.
Coverage stages	Values differ across reached lifecycle stages.	Validate $c_{\text{observation_resolved}}$ before application, then require exact $c_{\text{applied}} = c_{\text{observation_resolved}}$ at the event. Evaluate the inclusive predicates with c_{applied} ; create no future-stage placeholder and never require c_{realized} to authorize application.
Coverage order statistic	$N = 0$ and preregistered small populations around integer transitions.	Reproduce the exact zero-based $\lfloor ([0.75N] + N)/2 \rfloor$ index and selected ordered value; verify inclusive thresholds and finite-positive row requirement.
Bundle completeness	Required companions retained in memory, persisted, and consumed by a coadd-only route.	The logical bundle and plan remain complete; publication is required only for routes whose plan requires persistence.
Target grid	A preconstruction canonical grid is requested, or existing bundles have otherwise-compatible shapes.	Honor the requested grid; otherwise select a compatible common coadd target and use exact centered-integer embedding. Reject relabeling or resampling as a way to manufacture compatibility.
Crop	Explicit row selection after construction.	Emit a new identified product with selector/provenance; do not mutate or relabel the base grid.
Response	Valid fixed-state and procedure-response products.	Disclose exact MAP-local A_{MAP}, Π separately from any upstream or whole-chain claim; numerical procedure deltas require both exact maps into one authorized comparison domain and all gauge/support/perturbation facts.
Product roles	Independently request base, response-bearing, and covariance-qualified observation and coadd roles.	Preserve exact base parent and ordered membership; base-permitted unavailable companions do not reject; a selected required companion blocks only that stronger role and never forms a subset of an existing parent.
WCS population	Typed and FITS WCS differ at an interior center, outer boundary vertex, corner, and one invalid point in turn.	Evaluate the full preregistered population with exact spherical metric/frame/generation/ precision; reject on any invalid member and infer no continuum maximum.
NOI vocabulary	Present Q , formal precision, complete covariance, NOI conditional marginal second moment, NOI standardized signal, and an attempted significance label.	Preserve exact roles and claims; stronger physical-noise, uncertainty, significance, probability, false-rate, completeness, purity, or catalogue meaning remains unavailable without a separately approved method.

Pointing and OOF	Each requests exact ordinary MAP operator reuse.	Verify only the shared operator boundary; require separate mode-owned WCS/frame, model/fitting, interpretation, outputs, defaults, profile, and qualification, and leave an unsupplied mode profile unavailable.
Application status	Contract sources compile without an identified producer candidate.	Record contract completeness only; make no application availability or conformance claim.

SCI-NOI owns its exact realization-generation, empirical-uncertainty, conditional-scale, and standardization methods and products. Any physical- noise, total-uncertainty, covariance-completeness, significance, probability, false-rate, completeness, purity, or catalogue interpretation requires a separately approved exact method and remains unavailable by default. MAP therefore keeps Q normalization, formal conditional precision, complete conditional covariance, the NOI `conditional_detector_sign_randomization_marginal_second_moment`, the NOI standardized signal, and any future calibrated-significance method as distinct identities and claims.

5 Canonical requirement conformance clauses

The following clauses are generated from the shared authority. The first paragraph is the canonical science requirement; the second is its minimum conformance route. Every one of the 52 result fields is initialized to the exact value `not_assessed`; this says applicable evidence has not yet been executed or reviewed and does not describe artifact realization.

SCI-MAP-REQ-001 – Versioned authority

Canonical clause: The scientific contract version shall be v0.2. The rendered document revision shall be recorded separately as r0.2; changing layout or exposition alone shall not be represented as a new scientific contract version.

Minimum conformance evidence: Inspect both artifacts and their machine-readable provenance for distinct contract-version and document-revision fields.

Result: `not_assessed`

SCI-MAP-REQ-002 – Capability and realized PTC input

Canonical clause: The ordinary input shall be the realized SCI-PTC v0.1/r0.5 transformed calibrated- x product $z_i = Z_i^{\text{PTC}}$, a nonpolarimetric total-intensity-equivalent detector-time quantity in the inherited top-of-atmosphere, point-source-equivalent `mJy/beam` convention with its exact originating fixed-nominal-beam identity. Neither that unit nor a STOKES token establishes Stokes I. MAP has no CAL-to-MAP fallback, no inferred no-op PTC product, and no product when the required positive-rank PTC product is absent or PTC is disabled.

Minimum conformance evidence: Exercise exact realized PTC input, attempted CAL input, PTC-disabled and no-product routes, incompatible PTC generation, missing nominal-beam identity, and attempted Stokes-I promotion.

Result: `not_assessed`

SCI-MAP-REQ-003 – Exact occurrence identity

Canonical clause: Every candidate occurrence i shall bind observation, detector occurrence and UID, stable RTC output sample n , exact PTC product/application generation, segment, and array/network/group as applicable. Sample time is an associated attribute. Input-column, row, and container positions are locators only and shall not establish identity or equality.

Minimum conformance evidence: Permute times, containers, rows, and columns while preserving or breaking the exact parent chain; products shall follow exact occurrence identity and reject inferred joins.

Result: `not_assessed`

SCI-MAP-REQ-004 – Producer facts and MAP admission

Canonical clause: SCI-MAP shall consume PTC-owned transformed-signal availability, output-retention disposition, coefficient family/value/typed availability, and coefficient/QC proposition without reinterpretation. The PTC coefficient/QC proposition passes only when the coefficient is requested, applicable, eligible, and realized under the exact PTC-owned coefficient/QC profile for the declared MAP-facing family. It does not grant MAP use. SCI-MAP shall separately apply the source-bound occurrence profile `SCI-MAP:map_upstream_admission@2`; the manifest-bound SCI-VAL Registry revision binds and VAL Core evaluates that MAP-authored policy but does not author it. The immutable r0.5 profile `SCI-MAP:map_upstream_admission@1` is historical, is not a compatibility alias, and shall not replace `SCI-MAP:map_upstream_admission@2`. Every fact, decision, cause, scope, and immutable reason remains separately typed.

Minimum conformance evidence: Toggle each producer fact and each of the four named fields independently; verify exact PTC coefficient/QC projection, separate MAP admission, preserved causes, source/profile-version rejection, and no rescue or universal producer-flag action.

Result: `not_assessed`

SCI-MAP-REQ-005 – Coordinate and projection authority

Canonical clause: MAP shall consume `SCI-AST:rtc_output_grid_coordinates@1` for the same stable RTC sample n and complete parent chain as z_i ; it shall never join by time, row, cardinality, or coordinate equality. MAP owns G_{pi} through `SCI-MAP:one_hot_containing_pixel@1`: the unique finite in-grid pixel whose cell contains the coordinate under half-open bounds contributes $G_{pi} = 1$, every other pixel has $G_{pi} = 0$, and an outer-upper-boundary or out-of-grid coordinate contributes nowhere. Fractional projection is deferred. AST may materialize the relation only when the artifact binds the AST parent and exact MAP plan.

Minimum conformance evidence: Test exact-parent joins, centers, lower and upper cell boundaries, outer extent, invalid coordinates, sum-over-pixel normalization on its in-grid domain, and prohibited wrap/clamp/fractional placement.

Result: `not_assessed`

SCI-MAP-REQ-006 – Ordinary coefficient status

Canonical clause: The ordinary route requires one explicitly selected PTC-owned MAP-facing analysis/gridding coefficient family satisfying SCI-PTC-REQ-052–055 and SCI-PTC_TO_SCI-MAP_BOUNDARY v0.2-draft.1/r0.4. The admitted Registry SCI-PTC:analysis_gridding_coefficients@draft-0.1 contains SCI-PTC:uniform_constant@draft-0.1, whose exact supported value is dimensionless $\gamma_i = \omega_i = 1$, and grants MAP permission through SCI-PTC:uniform_coefficient_handoff@draft-0.1. Registration and permission do not select a family or realize a handoff. MAP shall use this family only after explicit selection; one immutable completed SCI-PTC v0.1/r0.5 transformed-product publication bound to the exact application, CAL parent, observation, and selection lineage and typed as a complete segment, complete scan, or bounded chunk wholly within one named scan or segment and one observation; exact target/payload/occurrence association; and a requested, applicable, eligible, realized profile evaluation under the admitted source and compatible binding generation. MAP does not reproduce, select, or infer the PTC-owned formula. An absent selection without an authorized mode default, unregistered family, incomplete PTC publication, missing or mismatched payload, unavailable coefficient/QC decision, or incompatible generation makes the affected route unavailable; there is no unity or alternate-family fallback. Structural availability remains distinct from finiteness. MAP alone classifies an authorized payload as finite positive, exact zero, negative finite, non-finite, or unrepresentable and applies only a matching finite strictly-positive value. The uniform family supplies no precision, sensitivity, loading, scatter, response, covariance, exposure, validity, optimality, or default meaning.

Minimum conformance evidence: Exercise the registered-but-unselected, selected-but-unpublished, missing/mismatched association, nonpassing four-axis evaluation, incompatible generation, and exact admitted uniform cases; verify producer QC, MAP admission, numerical classification, contribution, and every identity, support, lifecycle, uncertainty, and prohibited meaning separately.

Result: not_assessed

SCI-MAP-REQ-007 – Exposure lineage and accounting

Canonical clause: ALIGN-owned e_{acq} and valid-original e_{vo} , in seconds, shall remain immutable on stable original occurrences through RTC/CAL/PTC. The products are normatively **upstream_eligible_original_footprint_exposure** and **retained_original_footprint_exposure**. The shorter prose labels “upstream-eligible exposure” and “retained exposure” are explicit aliases for those products and no others. MAP shall deduplicate by exact observation, detector occurrence/UID, stable original/native occurrence, stable ALIGN slot, and ALIGN mapping generation; place each positive- e_{vo} original exactly once through SCI-AST_TO_SCI-MAP_ORIGINAL_FOOTPRINT_COORDINATE v0.1/r0.1, its layered AST ALIGN-grid direction/tangent/continuous-pixel parent in the exact target MAP WCS, and SCI-MAP:one_hot_containing_pixel@1; and never use a descendant RTC-output coordinate. Geometry/pointing realization, frame, coordinate validity, boundary state, parent, and version are required. Upstream-eligible and retained exposure differ only by ancestor-population restriction. Causal influence, representative RTC occurrence, and original-footprint exposure remain separate. These exposure products are not complete temporal support, effective map integration time, precision, or normalized-map influence. Missing original-coordinate authority makes the affected exposure role unavailable. Invalid-original has $e_{vo} = 0$; synthesis, replacement, donor reuse, overlapping filters, and decimation create no exposure. Boundary loss contributes nowhere. Coadd exposure unions atomically admitted observation-scoped original identities.

Minimum conformance evidence: Use one original influencing several descendants at different pixels, an original and descendants in different pixels, missing original coordinate, boundary loss, multiple maps sharing one original, distinct observations with identical coordinates, donor and synthesized descendants, and repeated lineage; verify exact own-coordinate placement, deduplication, and separate influence support.

Result: not_assessed

SCI-MAP-REQ-008 – Response information state

Canonical clause: Every valid SCI-MAP:base_observation_map@1 product shall disclose its known MAP-local realized operator response $A_{MAP,\Pi}$, including exact input/output domains, membership, projection, coefficient, support-row, WCS, parent, lifecycle, and limitations. MAP shall separately type upstream-conditioned fixed-state linear response, PTC full-procedure finite differences, and PTC+MAP re-resolved procedure response. An available fixed-state object shall also bind source domain, PTC-sample codomain, units, parents, response basis, and upstream limitations. Every numerical full-procedure Δz_{PTC-FP} , Δm_{PTC-FP} , or $\Delta m_{PTC+MAP-RR}$ shall bind the complete comparison-space record in A numerical full-procedure difference is defined only by a complete immutable comparison-space record. The record binds baseline and perturbed product identities; scientific quantity and unit; occurrence or MAP output-row identities; source and target frame/WCS; additive reference, gauge, and null- space state; baseline and perturbed support; coefficient and response basis; one exact common comparison domain; an exact map from each result into that domain; explicit treatment of rows present in only one result; perturbation amplitude, sign, side, location, source state, and normalization; finite- difference convention; state-transition record; and numerical availability with cause. It authorizes no implicit zero fill, union-domain sentinel, support intersection, coordinate-nearest or same-shape/row-position association, interpolation, reprojection, or gauge alignment. Without the complete authorized record the numerical difference is unavailable, or the result is only a separately typed state transition/discontinuity. ΔS_{PTC-FP} never establishes subtractability by itself.. The PTC+MAP family reruns PTC and separately resolves Π_0 and Π_+ , whose membership and S_{out} may differ; subtraction is unavailable without exact maps into one authorized common comparison

domain. A whole-chain RTC-to-CAL-to-PTC-to-MAP response is reserved for a separately authorized complete study and is unavailable unless so established. Absent upstream response shall be reported as limited or unavailable without concealing the known MAP-local operator, invalidating an otherwise valid base map, fabricating a whole-chain identity, or prohibiting later analysis. A later response estimate or corrected map is a new immutable versioned product bound to the original MAP product and processing identity.

Minimum conformance evidence: Exercise MAP-local-known/upstream-unavailable, fixed-state, full-procedure common-domain and unavailable-domain cases, PTC+MAP re-resolved changing membership, companion-starting-at-MAP, partial, and attempted whole-chain cases; verify exact typing, maps and rows, application exactly once, complete signal membership, honest limitations, and no fabricated identity or implicit subtraction.

Result: not_assessed

SCI-MAP-REQ-009 – Immutable plan and lifecycle

Canonical clause: The exact MAP lifecycle is **requested**, **effective**, **observation_resolved**, **applied**, **realized**, in that order. **requested** is immutable accepted intent. **effective** is its supported executable interpretation. **observation_resolved** means every observation-specific parent, coefficient, profile, geometry, mapping, parameter, support rule, output role, and persistence decision is concrete or typed unavailable. **applied** binds the exact resolved operation and parameter state to the invocation or commitment of one execution attempt. **realized** binds the immutable resulting accumulators, products, statuses, failures, and provenance that were actually produced. A stop before application has neither an applied nor a realized record. An applied attempt that terminates before output realization retains its applied-attempt identity and exact terminal failure but creates no realized product. The same ordered lifecycle shall bind the MAP plan, coefficient selection and application, projection, support policy, output plan, coadd plan, response and covariance requests, every independently requested product role, and **coverage_cut**. The MAP lifecycle does not replace the independently named PTC/VAL handoff axes **requested/not_requested**, **applicable/inapplicable/applicability_unknown**, **eligible/ineligible/decision_unavailable**, and **realized/incomplete/failed/not_produced**. It also does not replace the ECS requirement-result vocabulary or evidence-artifact realization state. In particular, a MAP parameter's resolved or applied value may be checked before application or bound after application; its **realized** value is never a prerequisite for a pre-application check. Disabled expert values remain inactive; one observation's resolved state shall not leak into another.

Minimum conformance evidence: Compare serialized states across two deliberately different observations; stop one before application and one after a terminal applied failure; verify ordered one-way provenance, absence of fabricated applied/realized products, exact applied-attempt failure, axis separation, and complete replacement.

Result: not_assessed

SCI-MAP-REQ-010 – Factored contribution predicate

Canonical clause: For each pixel, membership in C_p shall be the conjunction of ten exact Boolean pass predicates derived from separately retained typed records: transformed-signal availability; PTC output retention; coefficient availability; PTC coefficient/QC proposition pass, γ_{QC} ; MAP upstream admission; AST coordinate validity; finite z_i ; finite strictly positive γ_i ; admitted projection/boundary relation; and MAP-local contribution plus required-companion predicates. Every VAL-evaluated gate retains named request, applicability, eligibility, and realization fields; only request=**requested**, applicability=**applicable**, eligibility=**eligible**, and realization=**realized** passes. Every other field combination is nonpassing with its exact cause, failure scope, and immutable reason retained. A structural source- or profile-binding failure is recorded only when that structural condition actually occurs; an ordinary nonpassing field value is not thereby a structural failure. Evaluation follows the ordered DAG: A structural identity/parent, generation, source/profile, producer availability, and decision-realization checks; B retrieval only of structurally admitted payloads and MAP classification of z_i and γ_i ; C AST validity, one-hot boundary/placement, and MAP-local required companions; D membership construction; E accumulation. All gates pass before payload arithmetic. Reading an authorized payload to classify it is not arithmetic, and excluded non-finite values never contaminate accumulation. T/F/U/C or four-field records shall never be conjoined directly; no numeric-zero mask represents exclusion.

Minimum conformance evidence: Vary one gate, each of the four named fields, source-binding failure, and knowledge state at a time; compare the typed record, Boolean projection, exact DAG stage, membership, cause/scope, and immutable reason; place NaN/Inf behind structurally excluded and admitted cases to distinguish safe classification from arithmetic contamination.

Result: not_assessed

SCI-MAP-REQ-011 – Ordinary accumulators

Canonical clause: On the fixed contribution set, SCI-MAP shall accumulate $N_p = \sum_i a_{pi} z_i$, $Q_p = \sum_i a_{pi}$, and, when available for the identical membership, $K_p = \sum_i a_{pi} k_i$, where $a_{pi} = G_{pi} \gamma_i > 0$.

Minimum conformance evidence: Use one-pixel and one-hot hand calculations and compare accumulators before normalization; fractional projection is not authorized in v0.2.

Result: not_assessed

SCI-MAP-REQ-012 – Normalized publication

Canonical clause: The scientific normalized output vector shall contain exactly the rows $p \in \mathcal{S}_{\text{out}}$ authorized by the applicable effective support policy. On those rows finite $Q_p > 0$ gives $\hat{m}_p = N_p/Q_p$, and an available response companion gives $\hat{k}_p = K_p/Q_p$. No unsupported row, zero-filled value, or sentinel is a normalized scientific output or measured zero.

Minimum conformance evidence: Exercise positive, exact-zero, near-zero, overflowed, and non-finite aggregates; compare exact row identity and inspect any nonauthoritative storage slots.

Result: not_assessed

SCI-MAP-REQ-013 – Accumulator interpretation

Canonical clause: The numerator is a pre-normalized accumulator and Q_p is a normalization fact by default. Neither shall be labeled a sky estimator, inverse variance, confidence, coverage, or significance without the separately required conditions.

Minimum conformance evidence: Audit schema, metadata, API types, and labels for every numerator and normalization representation.

Result: not_assessed

SCI-MAP-REQ-014 – Constant preservation

Canonical clause: With fixed admitted membership and positive coefficients, the ordinary estimator shall preserve a constant admitted sample input on every row in its exact support-authorized output set. This prediction shall not create rows outside that set or be promoted to compact-source, extended-source, integrated-flux, absolute-offset, or astrometric fidelity.

Minimum conformance evidence: Map constant fixtures with uniform and unequal coefficients; compare exact output-row membership, then confirm that stronger response claims remain separately gated.

Result: not_assessed

SCI-MAP-REQ-015 – Projection-specific estimand

Canonical clause: The authorized base projection is exactly SCI-MAP:one_hot_containing_pixel@1; it preserves unity over the finite in-grid domain and has explicit half-open boundary loss. Fractional projection, diagonal weighted least squares, and conservation across a finite boundary are outside this branch until separately authorized.

Minimum conformance evidence: Use containing-pixel and outer-boundary fixtures; attempt fractional or WLS substitution and require method rejection.

Result: not_assessed

SCI-MAP-REQ-016 – One exact MAP operator for linear companions

Canonical clause: For one selected immutable realized plan Π , $A_{\text{MAP},\Pi} \equiv A_{\text{out}} \equiv D_{Q,\text{out}}^{-1} J_{\text{out}} G \Omega$ is the only MAP operator and its complete realized identity is the MAP-local operator-response disclosure. Its identity binds the exact admitted PTC-sample input domain, support-authorized output-row domain, coefficient family/generation, one-hot projection plan, support policy, target WCS/grid, immutable parents, and lifecycle generation. Signal, fixed-state response, conditional covariance, an available realized PTC-grid response companion, and fixed noise realizations shall use that same operator, membership, placement, coefficient state, and output rows. A PTC full-procedure difference with MAP state fixed may use the same operator only after its baseline and perturbed products, common PTC-sample comparison domain, exact input maps, common output-row comparison domain, and exact output maps are authorized; equality to $A_{\text{MAP},\Pi} \Delta z$ requires those exact common domains rather than array shape. A PTC-grid companion begins at MAP and receives the operator exactly once, never $H_{\text{fixed},\Theta}$ again. Physical original-footprint exposure instead uses unique originals and each original's own AST ALIGN-grid coordinate; causal influence is not a linear exposure companion.

Minimum conformance evidence: Trace all named uses coefficient by coefficient; compare operator, comparison-domain, row-map, and parent identities; verify complete MAP-local disclosure and exactly-once application; reject shape-, intersection-, interpolation-, and position-inferred differences; and separately trace original-footprint placement.

Result: not_assessed

SCI-MAP-REQ-017 – Realized response claim

Canonical clause: A stored kernel may be called a realized fixed-state response to template τ only when its exact sample parent satisfies $k = H_{\text{fixed},\Theta} \tau$ and the support-restricted map satisfies $\hat{k}_{\text{out}} = A_{\text{MAP},\Pi} H_{\text{fixed},\Theta} \tau = R_{\text{fixed},\Theta} \tau$ on exactly the signal rows. A PTC full-procedure finite difference $\Delta z_{\text{PTC-FP}}$ and either map difference shall retain its distinct family identity and the complete comparison-space record, including both product identities, quantity/unit, rows, frames/WCS, reference/gauge/null state, supports, bases, common domain and exact two maps, unmatched-row treatment, perturbation and finite-difference convention, state transition, availability, and cause. $\Delta S_{\text{PTC-FP}}$ records state change but never authorizes numerical subtraction. Otherwise the response is a tracer, separately typed discontinuity, limited, or unavailable.

Minimum conformance evidence: Inject a declared fixed-state template, comparable and noncomparable PTC full-

procedure perturbations, a PTC+MAP re-resolved perturbation with different Π and rows, and a mismatched tracer; verify family, complete comparison record, exact maps and rows, state differences, and no implicit fill, intersection, association, interpolation, reprojection, or gauge alignment.

Result: not_assessed

SCI-MAP-REQ-018 – Absolute-offset mode

Canonical clause: SCI-MAP v0.2 shall apply no signal-centering operation: $L = I$. Coaddition shall not subtract a mean, remove a constant mode, or recenter a source; any upstream loss of that mode shall remain explicit in response and provenance.

Minimum conformance evidence: Use a constant-offset fixture and inspect both the mapped value and recorded upstream response/null-mode state.

Result: not_assessed

SCI-MAP-REQ-019 – Conditional covariance

Canonical clause: When a declared PTC-domain covariance N_{PTC} is available for the exact fixed membership and state Θ , MAP shall propagate it as $C_{m,\text{out}} = A_{\text{MAP},\Pi} N_{\text{PTC}} A_{\text{MAP},\Pi}^T$, retaining consequential off-diagonal terms on the exact operator domains. The equation is complete only on the declared available domain. Missing, limited, symbolic, summarized, lineage-resolvable, or unavailable terms remain typed and are never zero-filled or interpreted as independence; absence of a complete model does not invalidate the signal map.

Minimum conformance evidence: Compare a small exact propagation and honest partial, symbolic, summarized, lineage-resolvable, and unavailable states on identical operator and row membership.

Result: not_assessed

SCI-MAP-REQ-020 – When normalization is precision

Canonical clause: Normalization may be called marginal inverse variance only when the exact selected γ family is calibrated true inverse variance, selected errors are independent on the asserted domain, and $\sum_i \gamma_i (G_{pi}^2 - G_{pi}) = 0$. One-hot geometry alone does not prove the coefficient premise.

Minimum conformance evidence: Test one-hot geometry with both proven inverse-variance and nonprecision coefficient families.

Result: not_assessed

SCI-MAP-REQ-021 – Formal diagonal weight

Canonical clause: When its assumptions hold, $\phi_p = Q_p^2 / \sum_i G_{pi}^2 \gamma_i$ is the formal marginal inverse variance. It remains distinct from Q_p , support-restricted precision $C_{m,\text{out}}^+$, exposure, hits, and empirical statistical weight.

Minimum conformance evidence: Compute and label every object for an authorized one-hot fixture and a correlated counterexample.

Result: not_assessed

SCI-MAP-REQ-022 – Covariance and nuisance disclosure

Canonical clause: The bundle shall report what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, lineage-resolvable, summarized, incomplete, or unavailable. It shall list omitted correlations and nuisance terms; absence of a complete covariance model shall not invalidate the map or prohibit later analysis, and unavailable information shall not be set to zero. A later covariance estimate may be attached only as a new versioned product bound to the original MAP product and processing identity.

Minimum conformance evidence: Inspect every representation state, prevent promotion of normalization, hits, or diagnostic weight to covariance, and verify derivative-product binding without rewriting the original claim.

Result: not_assessed

SCI-MAP-REQ-023 – Data-derived state

Canonical clause: If membership, coefficients, response, or geometry are estimated or selected from the same data, the displayed linear expectation and covariance shall remain explicitly conditional; unconditional bias or covariance requires a named joint model or resampling result outside this fixed-state claim.

Minimum conformance evidence: Use metadata cases for fixed and data-derived state and verify that unconditional labels appear only with separately admitted evidence.

Result: not_assessed

SCI-MAP-REQ-024 – No significance promotion

Canonical clause: Signal multiplied by the square root of merely formal weight shall be labeled formal standardized signal,

not `sig2noise` or empirical significance. SCI-NOI owns its exact realization-generation, empirical-uncertainty, conditional-scale, and standardization methods and products. Any physical- noise, total-uncertainty, covariance-completeness, significance, probability, false-rate, completeness, purity, or catalogue interpretation requires a separately approved exact method and remains unavailable by default. MAP therefore keeps Q normalization, formal conditional precision, complete conditional covariance, the NOI `conditional_detector_sign_randomization_marginal_second_moment`, the NOI standardized signal, and any future calibrated-significance method as distinct identities and claims. In particular, MAP shall not equate this quantity with NOI's conditional marginal second moment or standardized signal, and no future calibrated significance product is inferred.

Minimum conformance evidence: Attempt each prohibited relabeling through all public representations; compare Q , formal conditional precision, complete conditional covariance, the exact NOI marginal-second-moment and standardized-signal roles, and a hypothetical separately authorized significance role; require distinct identities and claims.

Result: `not_assessed`

SCI-MAP-REQ-025 – Distinct map facts

Canonical clause: Every base/unfiltered MAP bundle shall separately represent AST-geometric incidence, MAP-route-candidate incidence, estimator-admitted contribution, admitted-observation count, `upstream_eligible_original_footprint_exposure`, `retained_original_footprint_exposure`, numerical-normalization support, science-policy support, and MAP-local base-product scientific validity. No fact or downstream eligibility decision shall become authority for another.

Minimum conformance evidence: Construct cases in which each pair differs and verify separately typed identities and immutable ownership.

Result: `not_assessed`

SCI-MAP-REQ-026 – Incidence layers

Canonical clause: AST-geometric incidence records valid coordinate placement independent of signal availability; MAP-route-candidate incidence additionally requires the exact realized PTC route and occurrence-level upstream admission; estimator contribution additionally requires all remaining gates in $\mathcal{C}_p$. The one-hot touch convention counts the unique containing pixel only.

Minimum conformance evidence: Use coordinate-valid/signal-unavailable, route-ineligible, zero-coefficient, and admitted occurrences and compare all three populations.

Result: `not_assessed`

SCI-MAP-REQ-027 – Estimator contributions

Canonical clause: Estimator-admitted contributions shall count exactly $i \in \mathcal{C}_p$. A candidate ineligible under the exact MAP admission or contribution policy for its preserved cause, or with unavailable/conflicting required state, zero/negative/non-finite coefficient, non-finite signal, or out-of-grid projection, contributes nowhere; no producer flag implies a universal action.

Minimum conformance evidence: Compare exact counts with an independently evaluated profile and factored-gate oracle while preserving cause and decision scope.

Result: `not_assessed`

SCI-MAP-REQ-028 – Admitted observation count

Canonical clause: For coadds, admitted observation count shall count complete observation bundles contributing at the pixel and shall not be confused with sample hits or observation attempts.

Minimum conformance evidence: Use one complete, one unsupported, and one rejected observation bundle and verify the per-pixel count.

Result: `not_assessed`

SCI-MAP-REQ-029 – Upstream-eligible original-footprint exposure

Canonical clause: MAP `upstream_eligible_original_footprint_exposure` shall union and deduplicate exact positive- e_{vo} original occurrences that are ancestors of route candidates admitted by SCI-MAP:`map_upstream_admission@2`. Each original is placed once by its own exact AST coordinate and the one-hot half-open rule; it is not placed at every descendant coordinate. The exposure population precedes coefficient positivity and MAP-local numerical rejection. Missing original-coordinate or lineage authority makes this exposure role unavailable.

Minimum conformance evidence: Use one original influencing several candidates at different pixels, plus an eligible zero-coefficient candidate; verify one own-coordinate exposure placement, separate influence edges, and no precision promotion.

Result: `not_assessed`

SCI-MAP-REQ-030 – Retained original-footprint exposure

Canonical clause: MAP `retained_original_footprint_exposure` shall use the same original-occurrence key and own-coordinate placement as REQ-029, restricted to originals that are ancestors of estimator-admitted members. Overlapping filters, decimation, replacement, synthesis, and multiple descendants do not duplicate seconds. Coadded exposure shall union observation-scoped original identities after atomic coadd admission; one original can occupy at most one pixel under the base one-hot family, or none after boundary loss. Neither exposure is precision.

Minimum conformance evidence: Use overlapping observation lineage, multiple descendants, mixed admission, donor reuse, and boundary loss; compare exact original-key unions, pixel placement, and coadd sums.

Result: `not_assessed`

SCI-MAP-REQ-031 – Adopted threshold selector

Canonical clause: For each declared support-policy population, sort its finite strictly positive normalization coefficients Q_p ; for count $N > 0$, select zero-based $k = \lfloor ([0.75N] + N)/2 \rfloor$ and $Q_\star = Q_{(k)}$; for empty input set $Q_\star = 0$. The exact population and policy identities and each reached requested, effective, observation-resolved, applied, and realized state/value of dimensionless $c = \text{coverage_cut}$ shall be retained; a completed realization contains all five. Pre-application admission shall check the exact observation-resolved value. Application shall bind that value to one support-policy attempt; realization shall later bind only what was produced. The exact Boolean plan fact `coverage_cut_above_one_override_requested` shall bind request identity, authorized owner or profile, scope, each reached lifecycle stage, exact value, cause, and provenance. It has no applied/realized record before application and no realized record after an applied terminal failure. The scalar shall be finite and nonnegative. Exact zero requests no relative cut while finite strictly-positive Q_p remains mandatory. Values $0 < c \leq 1$ are the ordinary recommended range. A value $c > 1$ is admissible only when that independently authorized fact is true for the same resolved value and scope; the number cannot authorize itself and may honestly produce empty science support. Negative, non-finite, missing, or unrepresentable values shall fail before support-row construction, application, or required-product mutation. No universal numerical default is established.

Minimum conformance evidence: Check $N = 0$ and small N values around every integer-index transition against the exact zero-based formula; verify each reached stage and all five on completed realization, pre-application use of the resolved value, absence of future-stage placeholders, applied-terminal failure behavior, exact-zero and ordinary endpoints, independently authorized above-one override with same-value/scope matching, self-authorization rejection, possible empty support, and preconstruction nonmutation for every invalid state.

Result: `not_assessed`

SCI-MAP-REQ-032 – Separate support predicates

Canonical clause: Only after the effective plan and exact observation-resolved cut pass REQ-031 shall application bind $c_{\text{applied}} = c_{\text{observation_resolved}}$ for one support-policy attempt. Numerical normalization support shall then require finite positive $Q_p \geq Q_\star c_{\text{applied}}/10$, and science-policy support shall independently require finite positive $Q_p \geq Q_\star c_{\text{applied}}$. Both comparisons are inclusive and use the exact declared population and inherited order statistic. At $c = 0$, no relative threshold excludes a row, but no zero, negative, non-finite, or absent Q_p is restored. An authorized expert $c > 1$ may leave the science-support set empty. These thresholds are an operational relative-support policy and imply no exposure, precision, significance, response, completeness, or physical optimality. Neither support state establishes final validity or precision; no realized value is required to authorize this pre-application check.

Minimum conformance evidence: Exercise values below, exactly at, and above both thresholds for empty and nonempty populations; distinguish normalization from science support; verify resolved-to-applied equality, exact-zero behavior, no self-authorizing above-one value, and no use of c_{realized} as a prerequisite.

Result: `not_assessed`

SCI-MAP-REQ-033 – MAP-local base-product validity

Canonical clause: One MAP-local base-product scientific-validity state shall require membership in the exact support-authorized row set, exact admitted PTC and AST parents, passing MAP contribution/support policy, finite normalized signal, and absence of an applicable MAP-product failure. Explicitly limited or unavailable response/covariance does not alone invalidate the map. Retained storage, finite payload, coefficient, exposure, incidence, normalization, or downstream-local eligibility cannot establish or rewrite MAP validity.

Minimum conformance evidence: Use truth-table fixtures separating producer state, MAP admission, response/covariance disclosure, base validity, and later use-local eligibility.

Result: `not_assessed`

SCI-MAP-REQ-034 – Classify before arithmetic

Canonical clause: Only structurally admitted payloads shall be retrieved and safely classified for finiteness/sign before membership construction. All required gates shall pass before payload arithmetic. Multiplication by zero, masking after accumulation, or sanitizing a non-finite aggregate shall not substitute for admission.

Minimum conformance evidence: Place NaN and Inf payloads behind structural false and true predicates; verify no unauthorized read, exact numerical classification when authorized, no contamination, diagnostics, and failure scope.

Result: not_assessed

SCI-MAP-REQ-035 – Exceptional contribution states

Canonical clause: Zero coefficient shall be noncontributing, a low positive coefficient shall follow the explicit adopted policy, negative or non-finite ordinary coefficients shall be invalid, and an out-of-bounds projection shall contribute nowhere. Each cause shall remain distinguishable.

Minimum conformance evidence: Run one-cause-at-a-time fixtures and inspect cause-specific states plus unchanged neighboring pixels.

Result: not_assessed

SCI-MAP-REQ-036 – Complete observation bundle and roles

Canonical clause: The six independently requestable product roles are SCI-MAP:base_observation_map@01, SCI-MAP:response_bearing_observation_map@01, SCI-MAP:covariance_qualified_observation_map@01, SCI-MAP:base_coadd@01, SCI-MAP:response_bearing_coadd@01, and SCI-MAP:covariance_qualified_coadd@01. Every request has its own immutable identity, parent set, ordered MAP lifecycle, result, and failure scope. A stronger role is a new product over the exact base parent; it does not rewrite that parent's signal membership, response/covariance status, validity, or provenance. A SCI-MAP:base_observation_map@01 shall exist as one immutable ordered logical bundle before coaddition. It contains signal, numerator, normalization identity, complete MAP-local realized operator identity, honest upstream-response and uncertainty/covariance identity and status, the distinct facts in REQ-025, exposure lineage/accounting identity, unit and exact nominal-beam parent, complete AST frame/WCS, shape/indexing, estimator/projection/profile identities, PTC removed-component/additive-reference/null-space state, every actually reached stage of the five-stage MAP lifecycle, exact PTC/AST parentage, and product identity. It may remain valid when a stronger numerical upstream response or complete covariance is unavailable. Independently requested SCI-MAP:response_bearing_observation_map@01 and SCI-MAP:covariance_qualified_observation_map@01 products shall each parent the exact base observation, bind their own five-stage lifecycle, and require respectively every named numerical response object or every named within-observation covariance block. Physical persistence is separately controlled by the effective output plan.

Minimum conformance evidence: Request each observation role independently; remove or alter each required identity/companion; distinguish absent declarations from honest unavailable states; verify role-specific failure without base-parent mutation; and verify identical scientific content and identity for in-memory and faithfully persisted forms.

Result: not_assessed

SCI-MAP-REQ-037 – Role-qualified atomic coadd admission

Canonical clause: A complete observation bundle shall pass every compatibility check and only the companion checks required by the exact selected product role and SCI-MAP:observation_coadd_admission@02 before any coadd-owned state changes. Under SCI-MAP:base_coadd@01, unavailable or basis-incompatible response and incomplete covariance do not alter signal membership and do not reject an otherwise compatible base bundle; their exact identities, states, and causes remain. A required companion incompatibility is atomically decisive only for SCI-MAP:response_bearing_coadd@01 or SCI-MAP:covariance_qualified_coadd@01 as applicable. If an exact base coadd parent already exists, such an incompatibility blocks the stronger role for that parent and never selects a subset. A separately requested stronger-role admission may reject an observation only before constructing its own explicitly identified base/signal parent. Every rejection leaves all affected numerators, normalizations, counts, exposure, response, validity, provenance, and product cardinality unchanged.

Minimum conformance evidence: Snapshot all role-owned and base-parent state; inject one base fact, response, and covariance incompatibility at a time; verify role-specific admission, exact pre-mutation rejection, unchanged base membership, and no companion-only subset.

Result: not_assessed

SCI-MAP-REQ-038 – Centered integer common grid

Canonical clause: A canonical target grid may be requested and made immutable before observation-map construction. Observation maps constructed on that exact grid, or otherwise compatible observation/coadd shapes, shall use exact centered-integer embedding: shape differences are nonnegative and even in each axis, and the corresponding integer offset maps reference pixels to the same world coordinate. A later crop is a newly identified row-selected product with immutable parent and lineage; it shall not relabel the parent or make it retrospectively compatible. Fractional shift, post-hoc pad, reprojection, interpolation, wrapping, implicit recentering, mosaicking, and a different sky grid are outside this estimator.

Minimum conformance evidence: Test preconstruction target-grid requests, every parity/sign case and WCS reference-pixel relation, exact integer embedding, a separately identified post-hoc crop, attempted parent relabeling, and each excluded grid-changing operation.

Result: not_assessed

SCI-MAP-REQ-039 – Ordinary coadd estimator

Canonical clause: The exact v0.2 family `SCI-MAP:uniform_observation_coadd_coefficient@1`, owned by MAP, assigns dimensionless $u_{op} = 1$ to every row of each bundle eligible and realized under the VAL-governed aggregate profile `SCI-MAP:observation_coadd_admission@2`. The VAL request/applicability/eligibility/realization fields remain distinct from the five-stage MAP coadd lifecycle. The coefficient is an equal-observation coefficient, not precision or inverse variance. Each of `SCI-MAP:base_coadd@1`, `SCI-MAP:response_bearing_coadd@1`, and `SCI-MAP:covariance_qualified_coadd@1` is independently requestable, has its own exact base parent and lifecycle, and uses identical centered-integer equal-observation signal arithmetic for its admitted ordered membership. On exact row identities, $N_p^c = \sum_o u_{op} \hat{m}_{op}$, $Q_p^c = \sum_o u_{op}$, and $\hat{s}_p = N_p^c / Q_p^c$ is published only on $\mathcal{S}_{out}^c$. Conforming in-memory and persisted bundles have identical admission, identities, arithmetic, and result; storage route is not a scientific weighting or eligibility fact.

Minimum conformance evidence: Evaluate all four VAL fields separately from all MAP lifecycle stages; independently request all coadd roles; use one and two observations from in-memory and persisted conforming routes; verify exact parent/membership, arithmetic equality, and rejection of precision labels.

Result: `not_assessed`

SCI-MAP-REQ-040 – Coadd companions and unavailable response

Canonical clause: The stacked signal vector $\hat{\mathbf{m}}_{obs}$ shall carry explicit observation-row identities; B_{out} is the exact centered-integer equal-observation averaging operator; and T_{obs} is the stacked compatible fixed-state observation-response object on those same domains. `SCI-MAP:base_coadd@1` requires every admitted member's complete base bundle and all response identities/statuses, but permits unavailable or basis-incompatible response with unchanged numerical signal membership. It shall then publish coadd response unavailable with exact cause, never a hidden response subset, zero companion, or assumed compatibility. `SCI-MAP:response_bearing_coadd@1` parents one exact base coadd with identical ordered membership and requires every numerical response object named by the role with exact compatible source domain, basis, class, unit, normalization, parent, WCS/grid, and row identity. If the requirement fails for an existing parent, the stronger role is unavailable and the parent is unchanged. Any separately selected response-bearing admission that rejects a member shall construct a separately identified base/signal parent and reject before any of its state mutates. Exposure and count follow that selected admission, with exposure's unique-original placement.

Minimum conformance evidence: Use complete, basis-incompatible, limited, and unavailable responses under both roles; verify unchanged base membership and unavailable response, exact stronger-role parent identity, no subset of an existing parent, and atomic admission for a separately identified response-bearing request.

Result: `not_assessed`

SCI-MAP-REQ-041 – Coadd covariance and precision

Canonical clause: Σ^{obs} shall be the block covariance matrix on exact stacked observation-row identities with every available within- and cross-observation block explicitly located. `SCI-MAP:base_coadd@1` requires covariance identities/statuses and permits partial, symbolic, summarized, lineage-resolvable, or unavailable state. `SCI-MAP:covariance_qualified_coadd@1` parents one exact base coadd with identical ordered membership and requires every named within- and cross-observation block before MAP may publish $C_{c,out} = B_{out} \Sigma^{obs} B_{out}^T$. Missing blocks are unknown and block only this stronger role; they are never zero or independence. If a block is absent for an existing parent, no subset is selected and the base coadd remains unchanged. Any separately selected covariance-qualified admission that rejects a member shall construct its own separately identified base/signal parent before mutation. The uniform observation coefficient is not precision.

Minimum conformance evidence: Use complete within/cross blocks and every missing-block state under both roles; verify exact propagation only for the complete stronger role, unchanged base membership and validity, no subset, and exact parent/role identity.

Result: `not_assessed`

SCI-MAP-REQ-042 – Correlated-GLS boundary

Canonical clause: A correlated minimum-variance solution using $(\Sigma_p^{obs})^{-1} \mathbf{1}$ may require negative combination coefficients and therefore belongs to a separately authorized correlated-GLS branch, not the ordinary positive-coefficient coadd. The token K_p remains exclusively the MAP response/kernel accumulator and never names a covariance matrix.

Minimum conformance evidence: Use positive- and negative-correlation covariance matrices that distinguish ordinary weighting from the GLS benchmark; require method and notation identity to remain explicit.

Result: `not_assessed`

SCI-MAP-REQ-043 – Minimum product identity

Canonical clause: Every observation and coadd product shall identify one exact product-role token from the six-role Registry; its independent request and every actually reached stage of the five-stage MAP lifecycle; exact base parent and ordered membership; external observation/coadd, array/group, occurrence, and row domains; estimator and coefficient families; MAP-local operator and upstream-conditioned response/covariance states; signal unit and fixed-nominal-beam parent; complete AST

frame/WCS and WCS-generation identity; preconstruction target-grid request where applicable; grid/projection/boundary; MAP profiles; exact PTC product/application and coefficient generations; exposure convention; logical-bundle and plan-controlled persistence state; parentage; causes; and product identity. Any numerical full-procedure response additionally binds its complete comparison-space record; coverage policy binds its typed above-one override record. No STOKES token supplies quantity authority. Scientific identity shall be durable without prescribing a persistence format.

Minimum conformance evidence: Round-trip every role, lifecycle, parent, membership, comparison-space, and override identity through in-memory and persisted representations; detect collisions or generation mixing; and reject a bare token, storage route, or serialization convention as scientific authority.

Result: not_assessed

SCI-MAP-REQ-044 – Frames and units

Canonical clause: MAP shall bind the complete frame identity delivered by `SCI-AST:rtc_output_grid_coordinates@1`: exact source/target frame, reference system (including ICRS or FK5/J2000 as actually declared), equinox/epoch, time scale, site/model state, TAN topology, WCS, axis order, signs, handedness, and degree-valued science axes. MAP shall not equate J2000, FK5, ICRS, apparent, observed, or unrefracted frames by name or default. Signal/response units are `mJy/beam` with exact nominal-beam identity; covariance has its square.

Minimum conformance evidence: Inspect complete frame and unit/beam identity and require cross-frame, ambiguous-frame, cross-unit, or cross-beam coadd rejection.

Result: not_assessed

SCI-MAP-REQ-045 – WCS and index persistence

Canonical clause: Full-precision typed or sidecar WCS shall be the lossless admission and provenance authority. For an immutable finite MAP grid of n_x by n_y cells, the 0.1-arcsec representation-test population is the finite union of (a) every scientifically addressable integer pixel center $(j+1, k+1)$ for $0 \leq j < n_x$, $0 \leq k < n_y$, and (b) every outer-footprint cell vertex: $(h+\frac{1}{2}, \frac{1}{2})$, $(h+\frac{1}{2}, n_y+\frac{1}{2})$ for $0 \leq h \leq n_x$, together with $(\frac{1}{2}, l+\frac{1}{2})$ and $(n_x+\frac{1}{2}, l+\frac{1}{2})$ for $0 \leq l \leq n_y$. The second set includes all four outer corners and every required outer-boundary vertex. The complete population is preregistered from the full-precision typed WCS, grid shape, and row domain before the physical FITS WCS is generated. Both representations shall bind the same WCS-generation identity, celestial frame, axis order, signs, handedness, orientation, shape, and centered-integer reference-pixel relation. At every population member, each WCS is evaluated at the exactly corresponding center (zero-based memory index (j, k) , one-based FITS pixel $(j+1, k+1)$). Separation is the great-circle angle $\text{atan2}(\|\mathbf{u} \times \mathbf{v}\|, \mathbf{u} \cdot \mathbf{v})$ between unit direction vectors in that exact declared frame, evaluated with IEEE-754 binary64 round-to-nearest, ties-to-even operations and reported in arcseconds. The maximum over the complete finite population shall be at most 0.1 arcsec. If either WCS yields an invalid or non-finite direction at any preregistered member, the representation fails; the point is not dropped. A center is absent only when its scientific row was already absent before population generation; every listed outer-boundary vertex remains required. No smaller sampled subset or continuum/global-sky claim is authorized. The 0.1-arcsec bound is only the maximum over that exact preregistered finite population; it is not a continuum or global-sky maximum. Exact axis sign, handedness, orientation, and centered-integer shape/reference-pixel relations remain required independently of the numerical bound.

Minimum conformance evidence: Evaluate every preregistered center and outer-boundary vertex with the stated index relation, frame, WCS generation, binary64 spherical metric, and invalid-point rule; calculate the exact finite-population maximum and separately verify every orientation invariant.

Result: not_assessed

SCI-MAP-REQ-046 – Lifecycle-bound provenance

Canonical clause: Every MAP plan, coefficient selection/application, projection, support policy, output plan, coadd plan, response/covariance request, product role, and coverage policy shall record each reached requested, effective, observation-resolved, applied, and realized identity in order. The record shall include each reached exact `coverage_cut` stage, with all five included for completed realization; the typed `coverage_cut_above_one_override_requested` identity, authority/profile, scope, value, cause, provenance, and binding to every reached stage of the five-stage lifecycle; exact PTC/AST/boundary/profile and admitted coefficient Registry/family/handoff generations; every typed gate's independent request/applicability/eligibility/realization fields plus Boolean projection, immutable reason, cause, failure scope, and DAG stage; exact product-role and base-parent identities; admitted contributions/bundles; the complete MAP-local operator $A_{\text{MAP}, \Pi}$, accumulators, and row domains; unique-original exposure facts; PTC additive-reference/removed/null states; response family and complete comparison-space/state-transition record; covariance state; WCS comparison population/generation; centered-integer offsets; persistence state; required products; and failures. No applied or realized record shall be fabricated before application; a terminal post-application failure retains the applied attempt without a realized product.

Minimum conformance evidence: Reconstruct each reached lifecycle stage, role/parent, row-selected operator, cut/override transition, coefficient handoff, exposure population, gate projection, comparison-space map, WCS population, logical/persisted relation, and typed unavailable or terminal-failure state from provenance alone.

Result: not_assessed

SCI-MAP-REQ-047 – Failure scope and atomicity

Canonical clause: An invalid required input, unauthorized above-one cut, incompatible bundle, unrepresentable finite aggregate, unrepresentable coordinate/index conversion, missing companion required by the exact selected role, or failed required publication shall propagate at its declared contribution, pixel, observation-admission, or complete-product scope and shall fail before the affected live state changes. A pre-application failure creates no applied or realized record. A failure after application retains the exact applied attempt and terminal cause without fabricating a realized product or completion marker. Base-role response/covariance unavailability permitted by REQ-037–041 is an honest state, not a role-independent atomic rejection.

Minimum conformance evidence: Inject each failure class under every product role at its exact boundary; inspect scope, lifecycle, diagnostics, nonmutation, unchanged base parent, and absence of false application, realization, or completion.

Result: `not_assessed`

SCI-MAP-REQ-048 – Required observation products

Canonical clause: A complete logical `SCI-MAP:base_observation_map@1` bundle shall exist before coaddition. Persisted per-observation products are the normal scientific expectation and shall be published whenever the effective v0.2 output plan designates them required; every required publication shall complete or the declared operation fails without a completion claim. A plan may instead permit a fully in-memory route. A coadd-only entry may consume persisted conforming observation bundles without rerunning upstream processing. In-memory and persisted routes shall use identical logical admission, arithmetic, role and base-parent identity, five-stage lifecycle, and provenance. Optional nonpublication does not imply that the logical observation product did not exist, and this MAP rule does not waive PTC's upstream atomic required-publication obligations.

Minimum conformance evidence: Exercise required-publication success/failure, optional in-memory nonpublication, and coadd-only persisted input for each requested role; verify identical base bundle, parent, admission, arithmetic, lifecycle and provenance, exact failure propagation, no misleading realization/completion, and unchanged upstream PTC duties.

Result: `not_assessed`

SCI-MAP-REQ-049 – Immutable base parent

Canonical clause: Every stronger observation/coadd role and downstream transformation shall preserve the exact resolvable base/unfiltered MAP-bundle or base-coadd identity, ordered membership, lifecycle, and MAP-local base-product validity. A later local validity/eligibility decision may differ for its named use but shall not rewrite MAP validity, alter the original claims, sever parentage, or select a hidden subset of an existing base parent. Later response or covariance estimates are new versioned products.

Minimum conformance evidence: Create stronger roles and later locally eligible products over MAP-invalid and MAP-valid parents; verify exact membership, both validity states, parentage, and original claims remain immutable.

Result: `not_assessed`

SCI-MAP-REQ-050 – Complete consumer boundary

Canonical clause: No consumer shall receive a bare signal plane as a scientifically complete map. The boundary shall include exact product role, request, base parent and membership; quantity and nominal-beam identity; estimator/normalization/coefficient/profile identity; honest response class/state and uncertainty/covariance disclosure; distinct map and exposure facts; PTC additive-reference/removed-component/null-space state; exact frame/WCS, shape/indexing, all reached lifecycle stages, parentage, dependency versions, comparison-space record when applicable, and failures. Later analyses choose their own exact claim requirements and may attach honest versioned response/covariance products; MAP does not own their eligibility. SCI-NOI owns its exact realization-generation, empirical-uncertainty, conditional-scale, and standardization methods and products. Any physical- noise, total-uncertainty, covariance-completeness, significance, probability, false-rate, completeness, purity, or catalogue interpretation requires a separately approved exact method and remains unavailable by default. MAP therefore keeps Q normalization, formal conditional precision, complete conditional covariance, the NOI `conditional_detector_sign_randomization_marginal_second_moment`, the NOI standardized signal, and any future calibrated-significance method as distinct identities and claims.

Minimum conformance evidence: Exercise every role and claims with missing required facts, including NOI-related claims and analyses that explicitly accept limitations; verify fail-closed claim scope, exact parent/membership, and no rewrite of MAP validity.

Result: `not_assessed`

SCI-MAP-REQ-051 – Same fixed noise operator

Canonical clause: An admitted noise realization may be described as using the same map/coadd operator only when eligibility, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and atomic coadd admission are fixed or identically prescribed. Re-estimation from the randomized data defines a different named estimator.

Minimum conformance evidence: Propagate fixed deterministic vectors through signal and realization paths and compare row identities and operator coefficients, not filenames or shapes.

Result: not_assessed

SCI-MAP-REQ-052 – Method and claim separation

Canonical clause: Pointing and OOF may be registered consumers of the exact ordinary MAP gridding operator only when their inputs satisfy this operator contract. Reuse grants no mode-specific WCS/frame, source model, fitting, response interpretation, output, default, or qualification authority; an unsupplied mode profile remains unavailable. This authority shall not otherwise be applied by analogy to JINC, maximum-likelihood mapmaking, OOF residual transfer, reprojection, filtering, empirical noise, fitting, Beammap inference, feedback, Stokes Q/U, measured-R mapmaking, or other units. Independent inspection, modification, experimentation, and proposal remain unrestricted, while an altered scientific identity or official claim requires an explicit versioned successor and evidence appropriate to that claim. Algebraic contract correctness, engineering conformance, representation/response fidelity, observational performance, and production readiness shall remain separate claims.

Minimum conformance evidence: Exercise conforming and nonconforming Pointing/OOF operator requests, missing mode profiles, each prohibited method/claim promotion, and an independently changed support algorithm; verify exact operator reuse boundaries, unavailable unsupplied authority, versioned identity, and independent claim-state fields.

Result: not_assessed

6 Prediction execution matrix

Predictions are imported from the same shared authority. Engineering records attach fixtures, bounds, artifacts, and results; they do not rewrite the prediction.

SCI-MAP-PRED-001 – Uniform constant

Canonical prediction: For fixed membership, a constant admitted input with uniform positive coefficients produces that constant on every support-authorized MAP-valid output row; numerator and normalization scale together, and no unsupported scientific row is created.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-002 – Unequally weighted constant

Canonical prediction: For fixed membership, unequal positive coefficients still preserve the same constant on every support-authorized output row, while normalization and conditional variance generally change with position; rows outside that set have no normalized scientific value.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-003 – One-pixel hand calculation

Canonical prediction: For an authorized row with admitted z_i and $a_i > 0$, the exact triplet is $(N, Q, \hat{m}) = (\sum a_i z_i, \sum a_i, \sum a_i z_i / \sum a_i)$; an unauthorized row has no normalized scientific value.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-004 – One-hot boundary

Canonical prediction: A finite coordinate inside exactly one half-open cell has one $G_{pi} = 1$, preserves unity over pixels, and contributes only there. A coordinate on an internal upper edge belongs to the adjacent cell; a coordinate on the outer upper extent or outside the grid contributes nowhere. Fractional splitting is not authorized.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-005 – Delta input

Canonical prediction: For a declared fixed state, a basis-pixel sky perturbation produces the corresponding support-row-restricted column of $R_{\text{fixed}, \Theta} = A_{\text{MAP}, \Pi} H_{\text{fixed}, \Theta}$, including projection and boundary truncation. A PTC full-procedure or PTC+MAP re-resolved numerical difference exists only when its complete authorized comparison-space record supplies exact common domains and both row maps; absent that record it is unavailable or only a typed state transition/discontinuity and need not equal the fixed-state column.

Primary claim layer: Representation fidelity.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-006 – Point-source template

Canonical prediction: A declared point-source perturbation τ_b produces $R_{\text{fixed}, \Theta} \tau_b$ only for the exact fixed response state and authorized rows. PTC full-procedure and PTC+MAP re-resolved families additionally retain baseline/perturbed identities, perturbation side/sign/amplitude/location/source state/normalization, finite-difference convention, gauge/support/bases, exact common domain and row maps, state transition, availability, and cause. The unavailable whole-chain RTC-to-CAL-to-PTC-to-MAP family is not inferred. Peak, integrated, and fitted amplitudes remain distinct functionals.

Primary claim layer: Representation fidelity.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-007 – Resolved and Fourier inputs

Canonical prediction: At fixed state, resolved disks, gradients, and Fourier modes produce their explicit $R_{\text{fixed}, \Theta} \tau$ on the authorized row set; a constant-input pass does not predict preservation of these modes, and procedure-response families are not inferred from the fixed matrix.

Primary claim layer: Representation fidelity.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-008 – Variable coverage

Canonical prediction: Fixed-coefficient constant preservation can coexist with spatially varying normalization, support-row membership, covariance, and nonconstant-template response; a row that fails support is absent rather than measured zero.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-009 – Finite boundary

Canonical prediction: At an admitted finite edge, no contribution wraps or clamps; local normalization may preserve a constant while compact-source peak or integrated response changes.

Primary claim layer: Representation fidelity.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-010 – Coefficient states

Canonical prediction: The admitted uniform family yields exact dimensionless $\gamma_i = 1$ only for an explicit selection with complete PTC publication, exact target/payload/occurrence association, MAP permission, and a requested, applicable, eligible, realized named handoff under the exact compatible source binding. Registry presence alone realizes nothing. Missing selection or handoff remains unavailable with no fallback. Structural availability is independent of numerical class; MAP alone classifies finite positive, exact zero, negative finite, non-finite, and unrepresentable. Zero is noncontributing and the other nonpositive/non-finite classes are invalid with exact cause; uniform one is not precision.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-011 – Use-qualified exclusion and non-finite states

Canonical prediction: A candidate whose typed gate record projects to Boolean nonmembership does not contaminate an accumulator. Structural exclusion prevents payload retrieval; structurally admitted payloads are read only for safe MAP numerical classification before membership and arithmetic. Its complete named-field or other typed record, immutable reason, and cause remain preserved; no universal producer-flag action is inferred.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-012 – Threshold selector

Canonical prediction: For each predeclared positive finite coefficient population, the selected zero-based index is exactly $\lfloor ([0.75N] + N)/2 \rfloor$; empty input yields $Q_\star = 0$, but no row gains support without its own finite positive Q_p . Small N fixtures straddle every integer-index transition. At exact $c = 0$, both relative thresholds are zero and the finite-positive requirement still excludes zero, negative, non-finite, or absent normalization. Values $0 < c \leq 1$ use inclusive comparisons in the ordinary recommended range. A $c > 1$ value is admitted only with the independently true, same-value/scope `coverage_cut_above_one_override_requested` fact and may yield empty support; the number does not authorize itself. Negative, non-finite, missing, or unrepresentable c fails before application or required-product mutation. Every reached cut stage is retained, with all five retained for completed realization; pre-application validation checks the observation-resolved value and never requires a realized value.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-013 – Missing response

Canonical prediction: Every `SCI-MAP:base_observation_map@1` discloses its known MAP-local realized operator response on exact signal rows. Missing upstream response yields a separately typed limited or unavailable state on complete signal membership, never a smaller subset, zero, or identity. `SCI-MAP:response_bearing_observation_map@1` is unavailable for that exact base parent only when its exact selected role request names that upstream response object as required; absence of a different fixed-state, full-procedure, re-resolved, or whole-chain family does not block an otherwise complete named response role. Each numerical full-procedure difference requires its complete comparison-space record, and whole-chain response remains unavailable unless separately established. The base bundle remains usable for claims that do not require the absent response; any later response estimate or corrected map has a new versioned identity bound to the original.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-014 – Role-qualified atomic rejection

Canonical prediction: A required companion incompatibility rejects the complete observation before coadd mutation. Under `SCI-MAP:base_coadd@1`, unavailable or basis-incompatible response does not change signal membership; it yields unavailable coadd response with exact cause and no hidden subset or zero companion. Under a selected stronger role, an incompatible required response or covariance blocks that role for an existing base parent; any separately requested admission may reject the observation only before constructing its own explicitly identified base/signal parent. Other required identity, unit, WCS, shape, policy, and parent incompatibilities remain atomic at their declared role scope.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-015 – One-observation coadd

Canonical prediction: A `SCI-MAP:base_coadd@1` containing one admitted base observation under the uniform dimensionless coefficient family reproduces its signal on corresponding centered-integer authorized rows. Conforming in-memory and persisted bundles give the same admission, identity, and arithmetic. Unavailable or basis-incompatible upstream response leaves that signal membership unchanged and makes coadd response unavailable with exact cause. `SCI-MAP:response_bearing_coadd@1` reproduces response only when its exact base parent and every required response identity are compatible.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-016 – Uniform observation coefficients

Canonical prediction: On each authorized coadd row, two compatible observations produce their equal-observation arithmetic mean and count-like normalization under `SCI-MAP:uniform_observation_coadd_coefficient@1`. Each requested coadd role binds its own exact parent, lifecycle, and ordered membership; stronger companions do not change the arithmetic membership of their exact base parent. The normalization is not precision.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-017 – Missing or invalid observation

Canonical prediction: An unsupported observation row contributes at no coadd row; an observation rejected by the exact selected role contributes nowhere to that separately identified base/signal parent and does not increment its count. A companion unavailable but permitted by the base role does not remove the observation. Neither case creates a zero-valued scientific or companion substitute.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-018 – Centered integer placement

Canonical prediction: A canonical target grid requested before construction yields observation maps already bound to that grid. Other compatible shape differences that are nonnegative and even produce the exact half-difference row/column offset; odd, negative, fractional, or WCS-inconsistent relations fail with no mutation. The FITS representation comparison evaluates every preregistered addressable pixel center and outer-boundary cell vertex, including all corners, with the exact spherical metric and invalid-point rule. A later crop is a new row-selected product with its own identity and lineage and cannot relabel its parent as compatible.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-019 – Cross-observation covariance

Canonical prediction: For two maps with variances σ_1^2, σ_2^2 and covariance c , a complete `SCI-MAP:covariance_qualified_coadd@1` over the exact base-coadd membership includes c through Σ^{obs} . If a required block is missing, that stronger role is unavailable for the parent; the base role retains a partial, symbolic, summarized, lineage-resolvable, or unavailable state. No subset is formed, the block is never zero or independence, and equal-observation normalization is not precision.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-020 – Fixed noise operator

Canonical prediction: A deterministic admitted realization propagated with fixed state yields the same support-row identities and coefficient-by-coefficient $A_{\text{MAP},\Pi} \equiv A_{\text{out}}$ and B_{out} as the signal path; recomputed state exposes a named difference.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-021 – Sequential and parallel reduction

Canonical prediction: Integer facts, identities, masks, support, and cardinality agree exactly. Floating sums agree with a preregistered operation-count/conditioning bound that is frozen before results are viewed and is not loosened after failure.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-022 – No false significance

Canonical prediction: A deliberate attempt to label signal times square root of merely formal weight as empirical significance is rejected or remains unmistakably labeled formal standardized signal.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-023 – No validity promotion

Canonical prediction: A downstream product may have its own use-local validity or eligibility but cannot change MAP-local base-product validity or sever the base/unfiltered MAP parent.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-024 – No ordinary predicate for JINC

Canonical prediction: A deliberate attempt to apply positive-coefficient ordinary admission, support, or complete-bundle availability to JINC is rejected as outside SCI-MAP v0.2.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-025 – Aggregate and index failure

Canonical prediction: An exact-zero, non-finite, overflowed, or unrepresentable aggregate/index never becomes a live valid product; failure occurs at the declared scope before mutation or completion.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

7 Mechanical completeness checks

Before a conformance verdict, automated checks shall establish:

- exactly one declaration of each stable requirement and prediction ID in the shared authority, with no gaps or duplicates;
- every requirement appears in this specification and in `CROSSWALK.md`; every prediction is linked to at least one requirement;
- each rendered PDF identifies contract v0.2 and its current document revision separately;
- no unresolved LaTeX reference, citation, overfull box, clipped glyph, broken table, or unreadable page remains;
- every required artifact reopens successfully, has the expected page count, and renders every page through an independent PDF renderer; and
- the owner-decision ledger and author-draft decision record contain no silent resolution of an open scientific question.

The conformance verdict uses the same exact five-value requirement-result vocabulary. It is **pass** only when every applicable requirement is **pass** on the exact candidate. Any contradictory evaluated requirement makes it **fail**; otherwise any missing named prerequisite makes it **blocked**; otherwise any applicable unevaluated requirement makes it **not_assessed**. **not_applicable** is permitted only for a requirement outside the exact candidate's declared role with an owner-approved rationale. Evidence-artifact **incomplete** is never a requirement result. This document records every placeholder as **not_assessed** and asserts no verdict.

8 Required handoff contents

An engineering handoff contains the candidate and authority identities; requirement result table using exactly **pass/fail/ blocked/not_applicable/not_assessed**; a separate artifact-realization field using exactly **complete/incomplete/ failed/not_produced**; prediction result table; exact commands and environment; independent expected-value artifacts; build/test/log outputs; representation and observational evidence where claimed; failure-injection snapshots; skipped/blocked items; deviations and owner dispositions; and an explicit statement for each claim layer. The handoff shall not collapse “conforms algebraically,” “represents faithfully,” “performs on sky,” and “ready for production” into one status word.